THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ratio of 2 m 40 cm to 60 cm is: 1

(a) 4 : 2 **(b)** 5 : 1 **(c)** 1 : 4 **(d)** 4 : 1

2. If A : B = 2 : 3 and B : C = 6 : 5, then A : B : C is: 1

(a) 2 : 3 : 30 **(b)** 2 : 3 : 5 **(c)** 4 : 6 : 5 **(d)** 2 : 6 : 5

3. On a map, 1 cm represents 5 km. A distance of 3.5 cm on the map is actually: 1

(a) 22.5 km **(b)** 8.5 km **(c)** 17.5 km **(d)** 1.75 km

4. 8 workers can finish a job in 15 days. 12 workers will finish it in: 1

(a) 12 days **(b)** 22.5 days **(c)** 11 days **(d)** 10 days

5. A car covers 240 km in 4 hours at a steady speed. In 7 hours it covers: 1

(a) 420 km **(b)** 480 km **(c)** 67 km **(d)** 247 km

6. ₹ 1200 is divided among A, B and C in the ratio 1 : 3 : 4. The smallest share is: 1

(a) ₹ 600 **(b)** ₹ 150 **(c)** ₹ 240 **(d)** ₹ 450

7. If the number of workers increases and the job remains the same, the time taken: 1

(a) increases **(b)** decreases **(c)** stays the same **(d)** becomes zero

8. The ratio of 1 m 50 cm to 30 cm is: 1

(a) 1 : 5 **(b)** 5 : 1 **(c)** 5 : 2 **(d)** 6 : 1

9. If A : B = 3 : 4 and B : C = 8 : 7, then A : B : C is: 1

(a) 3 : 4 : 56 **(b)** 6 : 8 : 7 **(c)** 3 : 8 : 7 **(d)** 3 : 4 : 7

10. On a map, 1 cm represents 12 km. A distance of 4 cm on the map is actually: 1

(a) 4.8 km **(b)** 48 km **(c)** 60 km **(d)** 16 km

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the ratio of 3 m 60 cm to 90 cm in simplest form. 2

12. 9 workers complete a task in 16 days. How many days will 12 workers take? 2

13. A map scale is 1 cm = 20 km. Find the actual distance for 3.5 cm and the map distance for 100 km. 2

14. Find the ratio of 2 m 10 cm to 70 cm in simplest form. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If A : B = 2 : 3 and B : C = 6 : 5, find A : B : C. 3

16. 10 men can build a wall in 18 days. How many days will 15 men take to build it? 3

17. The ages of two friends are in the ratio 3 : 5 and the sum of their ages is 48 years. Find their ages after 4 years. 3

OR

If A : B = 3 : 4 and B : C = 8 : 7, find A : B : C. 3

18. 12 men can build a wall in 25 days. How many days will 20 men take to build it? 3

OR

The ages of two friends are in the ratio 2 : 3 and the sum of their ages is 60 years. Find their ages after 5 years. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Divide ₹ 1200 among A, B and C in the ratio 1 : 3 : 4. If C gives ₹ 100 to A, find the new ratio of their amounts. 5

OR

A bus covers 240 km in 4 hours at a uniform speed. Find 5
(i) its speed
(ii) the distance it covers in 7 hours
(iii) the time it takes to cover 360 km.

20. Divide ₹ 1800 among A, B and C in the ratio 2 : 3 : 4. If C gives ₹ 200 to A, find the new ratio of their amounts. 5

OR

A bus covers 180 km in 3 hours at a uniform speed. Find 5
(i) its speed
(ii) the distance it covers in 5 hours
(iii) the time it takes to cover 270 km.

END OF QUESTION PAPER — Total: 40 Marks

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